Travel Vaccination Advisory System

Step 1: Information Requirements

1.1 Travel Clinic Background

The Travel Vaccination Advisory System is a specialized healthcare application designed for a clinic operating within the travel medicine industry. This clinic serves as a dedicated "company" providing medical consultations and preventative care for individuals planning international travel. By operating at the intersection of healthcare and international travel, the clinic systematically matches a traveler's health profile against country-specific disease risks and vaccine requirements. Ultimately, the system automates the creation of professional advisory reports to ensure travelers receive accurate, data-driven medical guidance.

1.2 Database Problems and Scope

The clinic currently struggles with a manual and fragmented process for managing traveler information, leading to inefficiencies in providing timely health advice. Without a centralized system, it is difficult to accurately cross-reference a patient's immunization history with the specific health risks and vaccination requirements of their destination. This creates a risk of inconsistent medical recommendations and administrative errors. The scope of this project is to centralize traveler, trip, and clinical data to automate risk assessments and advisory report generation.

1.3 Business Operations and Data Requirements

Core Independent Entities

- **Traveler Enrollment**: The clinic collects personal details to create a permanent medical profile for every traveler. This includes the traveler's full name, date of birth, biological sex, phone number, and email address. Each traveler is assigned a unique identification number so their medical and travel history can be tracked over multiple years.

- **Country Directory**: The clinic maintains a directory of countries to evaluate destination-specific health risks. Each country is assigned a unique country code so it can be consistently referenced when reviewing travel plans, advisory rules, and disease case records.
- **Disease Reference**: The system manages a list of diseases that may pose risks to travelers. For each disease, the clinic stores descriptive information to help staff educate travelers about symptoms, transmission, and prevention. Each disease is assigned a unique identifier for tracking and reporting purposes.

Trip and Participation

- **Trip Planning**: When travelers schedule consultations, they provide details about upcoming travel, including travel dates and purpose (such as business, leisure, study, or volunteering). The system tracks the current status of each trip to reflect whether it is planned, in progress, completed, or cancelled. The status helps clinic staff understand whether a traveler is preparing for travel, currently traveling, or has already returned. Each travel plan is recorded as a distinct trip record so that medical advice can be tailored for that specific journey.
- **Trip Destinations (Country-GoingTo-Trip)**: Each trip is associated with exactly one destination country to ensure accurate evaluation of country-specific health risks and entry requirements. However, a single country may be the destination for many different trips over time.

- **Traveler Participation (Traveler-Takes-Trip)**: A traveler may participate in multiple trips over time. Likewise, a single trip may include multiple travelers, such as family members traveling together. The system must record which travelers are associated with which trips to ensure accurate documentation of group travel.

Medical Logic and Risk Data

- **Vaccine Protective Logic (Disease-Available-Vaccine)**: The clinic maintains a list of vaccines available for preventing diseases. Each vaccine is designed to prevent one specific disease, while a single disease may have multiple vaccines available. The clinic also tracks general vaccination coverage information for reference.
- **Geographic Advisory Rules (Country-Has-AdvisoryRule)**: The clinic establishes advisory rules based on destination country. Each advisory rule applies to exactly one country, but a country may have multiple advisory rules depending on different diseases or risk levels.
- **Disease-Specific Rules (AdvisoryRule-AppliesTo-Disease)**: Each advisory rule applies to one specific disease. However, a disease may have multiple advisory rules across different countries. These rules include a recommendation type (such as required or recommended) and a risk score to guide clinical decisions. Rules also include an effective date and notes so staff know when guidance applies and any special conditions.
- **Regional Disease Activity (Country-Has-ReportedCaseAndIncidence)**: The clinic tracks reported disease cases by country and year to assess current risk levels. A country may have multiple reported case records over different years and for different diseases.

- **Incidence Categorization (ReportedCaseAndIncidence-Gets-Disease)**: Each reported case record refers to one specific disease. However, a single disease may have multiple case records across various countries and years.

Consultation and Reporting

- **Trip Documentation (Trip-Has-AdvisoryReport)**: After reviewing the traveler's medical history and destination risks, the clinic generates exactly one formal health advisory report for each trip. Each advisory report is created for one specific trip and includes the date it was generated, summary notes, and the consultation fee for the services provided.
- **Report Contents (AdvisoryReport-Has-ReportItem)**: A single advisory report may contain multiple individual report items. Each report item belongs to exactly one advisory report and represents a specific medical recommendation or health instruction.
- **Vaccine Recommendations (ReportItem-Includes-Vaccine)**: Each report item includes exactly one vaccine recommendation along with a medical explanation of why it is required or recommended. However, a specific vaccine may appear in multiple report items across different advisory reports.

1.4 Translated Data Requirements and Examples

- **Traveler Entity**: The database needs to store information about each Traveler. For each traveler, store TravelerID, FName, LName, DOB, Sex, Phone, and Email. TravelerID is unique to each Traveler. Example: TR001 is James Miller, Male, born 05/14/1982, 920-627-1165, j.miller@example.com.

- **Trip Entity**: The database needs to store information about each Trip. For each trip, store TripID, StartDate, EndDate, TripStatus, and TravelPurpose. TripID is unique to each Trip. Example: TP001 starting 06/12/2024, ending 06/25/2024, status Completed, and purpose Study.
- **Country Entity**: The database needs to store information about each Country. For each country, store CountryId and CountryName. CountryId is unique to each Country. Example: C01 is Ghana.
- **Disease Entity**: The database needs to store information about each Disease. For each disease, store DiseaseId, DiseaseName, and Description. DiseaseId is unique to each Disease. Example: D02 is Measles and description is 'Highly contagious airborne viral infection caused by measles virus causing fever, cough, runny nose, red eyes, and a distinct red rash'.
- **Vaccine Entity**: The database needs to store information about each Vaccine. For each vaccine, store VaccineId, VaccineName, and VaccinationCoverage. VaccineId is unique to each Vaccine. Example: V01 is Measles containing vaccination covering 71%.
- **ReportedCaseAndIncidence Entity**: The database needs to store information about each ReportedCaseAndIncidence. For each record, store ReportedId, Year, and #ReportedCases. ReportedId is unique to each record. Example: ReportedId RC01 is for Year 2024 and recorded 450 #ReportedCases.
- **AdvisoryRule Entity**: The database needs to store information about each AdvisoryRule. For each rule, store RuleId, RiskScore, EffectiveDate, RecommendationType, and Notes. RuleId is unique to each rule. Example: RuleId R01 has a RiskScore of 5, an

EffectiveDate of 01/01/2026, a RecommendationType of "Required", and Notes stating "Required for border entry."

- **AdvisoryReport Entity**: The database needs to store information about each AdvisoryReport. For each report, store ReportId, SummaryNotes, GeneratedDate, and ConsultationFee. ReportId is unique to each report. Example: ReportId AR01 contains SummaryNotes of "Patient is fit for travel," a GeneratedDate of 02/20/2026, and a ConsultationFee of 125.
- **ReportItem Entity**: The database needs to store information about each ReportItem. For each item, store ReportItemId, RecommendationType, and Reason. ReportItemId is unique to each item. Example: ReportItemId RI01 has a RecommendationType of "Required" and a Reason of " Measles risk in crowded travel settings; ensure immunity "
- **Traveler–Takes–Trip Relationship**: The database needs to store which travelers participate in which trips. Each traveler can participate in multiple trips. Each trip can include multiple travelers. Example: Traveler TR001 participates in Trips TP001 and TP003. Trip TP001 includes Travelers TR001, TR002, TR003, and TR004.
- **Country–GoingTo–Trip Relationship**: The database needs to store the destination Country for each Trip. Each country can be the destination for many trips. Each trip is associated with exactly one country. Example: Country C01 (Ghana) is the destination for Trips TP001 and TP003. Trip TP001 is going to Country C01 (Ghana).
- **Disease–Available–Vaccine Relationship**: The database needs to store which Vaccines are available for each Disease. Each vaccine prevents only one disease. Each disease can be prevented by multiple vaccines. Example: Vaccine V01 prevents Disease D02 (Measles). Disease D02 (Measles) is associated with Vaccine V01 and V06.

- **Country–Has–AdvisoryRule Relationship**: The database needs to store which AdvisoryRules apply to each Country. Each country can have multiple advisory rules. Each advisory rule applies to exactly one country. Example: Country C01 (Ghana) has AdvisoryRules R01 and R02. AdvisoryRule R01 applies to Country C01 (Ghana).
- **AdvisoryRule–AppliesTo–Disease Relationship**: The database needs to store which Disease each AdvisoryRule applies to. Each advisory rule applies to exactly one disease. Each disease can have multiple advisory rules. Example: AdvisoryRule R01 applies to Disease D02 (Measles). Disease D02 (Measles) has AdvisoryRules R01 and R07.
- **Trip–Has–AdvisoryReport Relationship**: The database needs to store which AdvisoryReport is generated for each Trip. Each trip has exactly one advisory report. Each advisory report is generated for exactly one trip. Example: Trip TP001 has AdvisoryReport AR01. AdvisoryReport AR01 is generated for Trip TP001.
- **AdvisoryReport–Has–ReportItem Relationship**: The database needs to store which ReportItems belong to each AdvisoryReport. Each advisory report can contain multiple report items. Each report item belongs to exactly one advisory report. Example: AdvisoryReport AR01 contains ReportItems RI01 and RI02. ReportItem RI01 belongs to AdvisoryReport AR01.
- **ReportItem-Includes-Vaccine Relationship:** The database needs to store which Vaccine is included in each ReportItem. Each ReportItem includes exactly one Vaccine. Each Vaccine can be included in multiple ReportItems. Example: ReportItem RI01 includes Vaccine V01. Vaccine V01 appears in ReportItems RI01 and RI04.
- **Country–Has–ReportedCaseAndIncidence:** The database needs to store ReportedCaseAndIncidence records for each Country. Each Country can have multiple

ReportedCaseAndIncidence records (across diseases and years). Each ReportedCaseAndIncidence record belongs to exactly one Country. Example: Country C02 (Kenya) has ReportedCase records RC02 and RC05. Record RC05 belongs to Country C02 (Kenya).

- **ReportedCaseAndIncidence–Gets–Disease:** The database needs to store which Disease each ReportedCaseAndIncidence record refers to. Each ReportedCaseAndIncidence record refers to exactly one Disease. Each Disease can have multiple ReportedCaseAndIncidence records (across countries/years). Example: Record RC05 refers to Disease D02 (Measles). Disease D03 (Rubella) appears in records RC01 and RC02.

1.5 Sample Data

Traveler Intake

Household / Contact	Traveler Full Name	DOB	Sex	Email	Phone
Miller Family	James Miller	05/14/1982	Male	j.miller@email.com	555-010-0101
Miller Family	Sarah Miller	08/22/1984	Female	s.miller@email.com	555-010-0101
Miller Family	Leo Miller	11/30/2012	Male	kid1.miller@email.com	555-010-0101
Miller Family	Mia Miller	03/12/2015	Female	kid2.miller@email.com	555-010-0101
Wilson Household	Robert Wilson	01/15/1978	Male	r.wilson@email.com	555-020-0202

Trip Request

Traveler / Group	Destination Country	Start Date	End Date	Purpose	Status	Notes
Miller Family	Ghana	06/12/2024	06/25/2024	Leisure	Completed	Family trip
Robert Wilson	Kenya	05/12/2025	05/22/2025	Volunteer	Completed	NGO site visit

Planned Traveler	Ghana	09/05/2026	09/20/2026	Work	Planned	Business meetings
Student Traveler	Thailand	01/15/2026	02/05/2026	Study	Cancelled	Schedule change

Trip Roster

Trip Description	Traveler Full Name	DOB	Notes
Ghana Leisure Trip (06/12/2024–06/25/2024)	James Miller	05/14/1982	Adult
Ghana Leisure Trip (06/12/2024–06/25/2024)	Sarah Miller	08/22/1984	Adult
Ghana Leisure Trip (06/12/2024–06/25/2024)	Leo Miller	11/30/2012	Child
Ghana Leisure Trip (06/12/2024–06/25/2024)	Mia Miller	03/12/2015	Child
Kenya Volunteer Trip (05/12/2025–05/22/2025)	Robert Wilson	01/15/1978	Solo

Country Directory

Country Name	Region	Notes
Ghana	Africa	Vaccine guidance varies by traveler history
Kenya	Africa	Volunteer travel may increase exposure risks
Thailand	Asia	Consider outbreak notices seasonally
Vietnam	Asia	Verify routine immunizations

Disease Reference

Disease Name	Description
Diphtheria	Highly contagious bacterial infection affecting nose, throat, skin
Measles	Highly contagious airborne viral infection causing fever and rash
Rubella	Viral infection with mild rash/fever; serious in pregnancy
Mumps	Viral infection causing swelling of salivary glands

Vaccine Reference

Vaccine Name	Coverage	Protects Against

Measles containing vaccination	71%	Measles
Rubella containing vaccination	37%	Rubella
DTP-containing vaccine	89%	Diphtheria

Advisory Rules

Destination Country	Disease	Required/Recommended	Risk Score (1-5)	Effective Date	Notes
Ghana	Measles	Required	5	01/01/2026	Proof required for school/work entry in some regions
Ghana	Diphtheria	Recommended	4	01/01/2026	Booster advised if last dose > 10 years
Thailand	Rubella	Recommended	3	01/01/2026	Review immunity before traveling to crowded areas
Kenya	Mumps	Recommended	3	01/01/2026	Increased risk in outbreaks; confirm vaccination

Disease Activity Log

Country	Disease	Year	Reported Cases	Source / Notes
Ghana	Rubella	2024	31593	Public health summary
Kenya	Rubella	2019	22	Historical record
Thailand	Mumps	2024	121	Public health summary
Vietnam	Diphtheria	2022	62	Public health summary
Kenya	Measles	2023	669083	Public health summary

Advisory Report

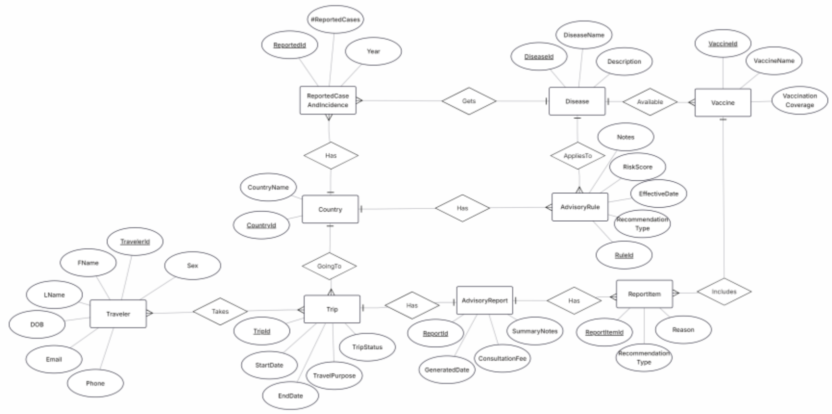
Traveler/Group	Destination	Travel Dates	Date	Consultation Fee	Summary Notes
Miller Family	Ghana	06/12/2024–06/25/2024	06/01/2024	$75	Reviewed itinerary and vaccine history;

					recommendations provided
Robert Wilson	Kenya	05/12/2025–05/22/2025	05/01/2024	$60	Discussed routine immunizations and travel precautions
Planned Traveler	Ghana	09/05/2026–09/20/2026	08/25/2024	$80	Reviewed outbreak concerns and vaccination gaps

Advisory Report Items

Traveler/Group	Destination	Travel Dates	Date	Recommendation Type	Recommended Vaccine	Reason / Notes
Miller Family	Ghana	06/12/2024–06/25/2024	06/01/2024	Required	Measles containing vaccination	Measles risk in crowded travel settings; ensure immunity before travel.
Miller Family	Ghana	06/12/2024–06/25/2024	06/01/2024	Recommended	Rubella containing vaccination	Consider rubella protection based on vaccination history.
Planned Traveler	Ghana	09/05/2026–09/20/2026	08/25/2024	Recommended	Measles containing vaccination	Measles immunity is recommended due to outbreak risk and work travel.

Step 2: E-R Diagram



Explanation of Entities and Relationships

- Traveler: A traveler is a person who receives travel health consultation from the clinic.
- Trip: A trip represents a specific travel plan with defined travel dates, purpose, and destination.
- Country: A country is a travel destination linked to trips, advisory rules, and reported disease cases.
- ReportedCaseAndIncidence: This stores the number of reported cases of a disease in a country for a specific year.
- Disease: A disease is a health condition that may present a risk to travelers.
- Vaccine: A vaccine is an immunization that prevents a specific disease.
- AdvisoryRule: An advisory rule defines the clinic’s recommendation for a specific disease in a specific country.
- AdvisoryReport: An advisory report is the consultation document created for a trip.
- ReportItem: A report item is an individual recommendation included in an advisory report.
- Traveler–Takes–Trip: Records which travelers participate in which trips.

- Country–GoingTo–Trip: Records the destination country for each trip.
- Trip–Has–AdvisoryReport: Records which advisory report is created for each trip.
- AdvisoryReport–Has–ReportItem: Records which report items belong to an advisory report.
- ReportItem–Includes–Vaccine: Records which vaccine is included in a report item.
- Country–Has–AdvisoryRule: Records which advisory rules apply to a country.
- AdvisoryRule–AppliesTo–Disease: Records which disease an advisory rule applies to.
- Country–Has–ReportedCaseAndIncidence: Records reported disease case data for a country.
- ReportedCaseAndIncidence–Gets–Disease: Records which disease each case record refers to.
- Disease–Available–Vaccine: Records which vaccines are available for a disease.

Step 3: Converting to tables

A) Convert all the tables

Traveler: (<u>TravelerId</u>, FName, LName, DOB, Sex, Email, Phone)

Trip: (<u>TripId</u>, StartDate, EndDate, TripStatus, TravelPurpose)

Country: (<u>CountryId</u>, CountryName)

ReportedCaseAndIncidence: (<u>ReportedId</u>, Year, #ReportedCases)

Disease: (<u>DiseaseId</u>, DiseaseName, Description)

Vaccine: (<u>VaccineId</u>, VaccineName, VaccinationCoverage)

AdvisoryRule: (<u>RuleId</u>, RiskScore, EffectiveDate, RecommendationType, Notes)

AdvisoryReport: (<u>ReportId</u>, SummaryNotes, GeneratedDate, ConsultationFee)

ReportItem: (<u>ReportItemId</u>, RecommendationType, Reason)

B) convert all relationships

Traveler-Takes-Trip Relationship: This is a n-m relationship. So make a new table TraveleronTrip:

TraveleronTrip: (*<u>TripId</u>*, *<u>TravelerId</u>*) — Foreign keys: *TravelerId, TripId*

Country-GoingTo-Trip Relationship: This is a 1-n relationship. Add the primary key of "1" entity (Country) to the many side (Trip) as a foreign key.

Trip: (<u>TripId</u>, StartDate, EndDate, TripStatus, TravelPurpose, *CountryId)* — Foreign key: *CountryId*

Trip-Has-AdvisoryReport Relationship: This is a 1-1 relationship: Since each trip may not yet have a corresponding advisory report, add TripId (the "1" side) to AdvisoryReport (pretend "many" side) to avoid null values.

AdvisoryReport: (<u>ReportId</u>, SummaryNotes, GeneratedDate, ConsultationFee, *TripId*) — Foreign key: *TripId*

AdvisoryReport-Has-ReportItem Relationship: This is a 1-n relationship . Add the primary key of "1" entity (AdvisoryReport) to the many side (ReportItem) as a foreign key.

ReportItem-Includes-Vaccine Relationship: This is a n-1 relationship . Add the primary key of "1" entity (Vaccine) to the many side (ReportItem) as a foreign key.

ReportItem: (<u>ReportItemId</u>, RecommendationType, Reason, *ReportId, VaccineId*) — Foreign keys: *ReportId, VaccineId*

Country-Has-AdvisoryRule Relationship: This is a 1-n relationship. Add the primary key of "1" entity (Country) to the many side (AdvisoryRule) as a foreign key.

AdvisoryRule-AppliesTo-Disease Relationship: This is a n-1 relationship. Add the primary key of "1" entity (Disease) to the many side (AdvisoryRule) as a foreign key.

AdvisoryRule: (RuleId, RiskScore, EffectiveDate, RecommendationType, Notes, *CountryId*, *DiseaseId*) — Foreign keys: *CountryId, DiseaseId*

Country-Has-ReportedCaseAndIncidence Relationship: This is a 1-n relationship. Add the primary key of "1" entity (Country) to the many side (ReportedCaseAndIncidence) as a foreign key.

ReportedCaseAndIncidence-Gets-Disease Relationship: This is a n-1 relationship . Each incidence record is for one disease; one disease can have many records. Add the primary key of "1" entity (Disease) to the many side (ReportedCaseAndIncidence) as a foreign key.

ReportedCaseAndIncidence: (ReportedId, Year, #ReportedCases, *CountryId, DiseaseId*) — Foreign keys: *CountryId, DiseaseId*

Disease-Available-Vaccine Relationship: This is a 1-n relationship. Add the primary key of "1" entity (Disease) to the many side (Vaccine) as a foreign key.

Vaccine: (VaccineId, VaccineName, VaccinationCoverage, *DiseaseId*) — Foreign key: *DiseaseId*

C) Final Conversion Result

Traveler: (TravelerId, FName, LName, DOB, Sex, Email, Phone)

TraveleronTrip: (*TripId*, *TravelerId*) — Foreign keys: *TravelerId, TripId*

Trip: (TripId, StartDate, EndDate, TripStatus, TravelPurpose, *CountryId*) Foreign keys: *CountryId*

Country: (CountryId, CountryName)

ReportedCaseAndIncidence: (ReportedId, Year, #ReportedCases, *CountryId, DiseaseId*) Foreign keys: *CountryId, DiseaseId*

Disease: (DiseaseId, DiseaseName, Description)

Vaccine: (VaccineId, VaccineName, VaccinationCoverage, *DiseaseId)* Foreign key: *DiseaseId*

AdvisoryRule: (RuleId, RiskScore, EffectiveDate, RecommendationType, Notes, *CountryId, DiseaseId*) Foreign keys: *CountryId, DiseaseId*

AdvisoryReport: (ReportId, SummaryNotes, GeneratedDate, ConsultationFee, *TripId*) Foreign key: *TripId*

ReportItem: (ReportItemId, RecommendationType, Reason, *ReportId, VaccineId*) Foreign keys: *ReportId, VaccineId*

D) Functional Dependencies Check and Normal Form Determinations

Traveler: (TravelerId, FName, LName, DOB, Sex, Email, Phone) FD: TravelerId → FName, LName, DOB, Sex, Email, Phone. No additional FDs. In BCNF.

TraveleronTrip: (*TripId*, *TravlerId*) No FD. In BCNF

Trip: (TripId, StartDate, EndDate, TripStatus, TravelPurpose, *CountryId)* FD: TripId → StartDate, EndDate, TripStatus, TravelPurpose, CountryId. No additional FDs. In BCNF.

Country: (CountryId, CountryName) FD: CountryId → CountryName. No additional FDs. In BCNF.

ReportedCaseAndIncidence: (ReportedId, Year, #ReportedCases, *CountryId, DiseaseId)* FD: ReportedId → Year, #ReportedCases, CountryId, DiseaseId. No additional FDs. In BCNF.

Disease: (DiseaseId, DiseaseName, Description) FD: DiseaseId → DiseaseName, Description. No additional FDs. In BCNF.

Vaccine: (VaccineId, VaccineName, VaccinationCoverage, *DiseaseId)* FD: VaccineId → VaccineName, VaccinationCoverage, DiseaseId. No additional FDs. In BCNF.

AdvisoryRule: (RuleId, RiskScore, EffectiveDate, RecommendationType, Notes, *CountryId, DiseaseId*) FD: RuleId → RiskScore, EffectiveDate, RecommendationType, Notes, CountryId, DiseaseId. No additional FDs. In BCNF.

AdvisoryReport: (ReportId, SummaryNotes, GeneratedDate, ConsultationFee, TripId) FD: ReportId → SummaryNotes, ConsultationFee, GeneratedDate, TripId. TripId → ReportId. Table is in BCNF because TripId is a candidate key.

ReportItem: (ReportItemId, RecommendationType, Reason, *ReportId, VaccineId*) FD: ReportItemId → RecommendationType, Reason, ReportId, VaccineId. No additional FDs. In BCNF.

E) Final Result after Normalization

Traveler: (TravelerId, FName, LName, DOB, Sex, Email, Phone)

Trip: (TripId, StartDate, EndDate, TripStatus, TravelPurpose, *CountryId*) Foreign keys: *CountryId*

TraveleronTrip: (*TripId*, *TravelerId*) — Foreign keys: TravelerId, TripId

Country: (CountryId, CountryName)

ReportedCaseAndIncidence: (ReportedId, Year, #ReportedCases, *CountryId, DiseaseId*) Foreign keys: *CountryId, DiseaseId*

Disease: (DiseaseId, DiseaseName, Description)

Vaccine: (VaccineId, VaccineName, VaccinationCoverage, *DiseaseId)* Foreign key: *DiseaseId*

AdvisoryRule: (RuleId, RiskScore, EffectiveDate, RecommendationType, Notes, *CountryId, DiseaseId*) Foreign keys: *CountryId, DiseaseId*

AdvisoryReport: (ReportId, SummaryNotes, GeneratedDate, ConsultationFee, *TripId*) Foreign key: *TripId*

ReportItem: (ReportItemId, RecommendationType, Reason, *ReportId, VaccineId*) Foreign keys: *ReportId, VaccineId*

F) Check foreign key

There are 10 tables with 11 foreign keys. Since the number of foreign keys (11) >= number of tables - 1 (9), this satisfies the minimum number of foreign keys required.

Step 4) Final database tables with data

Traveler: (TravelerId, FName, LName, DOB, Sex, Email, Phone)

TravelerId	FName	LName	DOB	Sex	Email	Phone
TR001	James	Miller	05/14/1982	Male	j.miller@email.com	555-010-0101
TR002	Sarah	Miller	08/22/1984	Female	s.miller@email.com	555-010-0101
TR003	Leo	Miller	11/30/2012	Male	kid1.miller@email.com	555-010-0101
TR004	Mia	Miller	03/12/2015	Female	kid2.miller@email.com	555-010-0101
TR005	Robert	Wilson	01/15/1978	Male	r.wilson@email.com	555-020-0202

Trip: (TripId, StartDate, EndDate, TripStatus, TravelPurpose, *CountryId*) **Foreign Key: CountryId**

TripId	StartDate	EndDate	TripStatus	TravelPurpose	CountryId
TP001	06/12/2024	06/25/2024	Completed	Leisure	C01
TP002	05/12/2025	05/22/2025	Completed	Volunteer	C02
TP003	09/05/2026	09/20/2026	Planned	Work	C01
TP004	01/15/2026	02/05/2026	Cancelled	Study	C03

TraveleronTrip: (*TravelerId, TripId*) **Foreign Keys: TravelerId, TripId**

TripId	TravelerId
TP001	TR001
TP001	TR002
TP001	TR003
TP001	TR004
TP002	TR005

Country: (CountryId, CountryName)

CountryId	CountryName
C01	Ghana
C02	Kenya

C03	Thailand
C04	Vietnam

ReportedCaseAndIncidence: (ReportedId, Year, #ReportedCases, *CountryId, DiseaseId*) **Foreign Keys: CountryId, DiseaseId**

ReportedId	Year	#ReportedCases	CountryId	DiseaseId
RC01	2024	31593	C01	D03
RC02	2019	22	C02	D03
RC03	2024	121	C03	D04
RC04	2022	62	C04	D01
RC05	2023	669083	C02	D02

Disease: (DiseaseId, DiseaseName, Description)

DiseaseId	DiseaseName	Description
D01	Diphtheria	Highly contagious bacterial infection caused by Corynebacterium diphtheriae, primarily affecting the nose, throat, and skin
D02	Measles	Highly contagious airborne viral infection caused by measles virus causing fever, cough, runny nose, red eyes, and a distinct red rash
D03	Rubella	A contagious viral infection caused by Rubivirus causing mild symptoms like a rash starting on the face, low-grade fever, and swollen glands
D04	Mumps	A contagious viral infection caused by the mumps virus, characterized by painful swelling of the salivary (parotid) glands

Vaccine: (VaccineId, VaccineName, VaccinationCoverage, *DiseaseId*) **Foreign Key: DiseaseId**

VaccineId	VaccineName	VaccinationCoverage	DiseaseId
V01	Measles containing vaccination	71%	D02
V02	Rubella containing vaccination	37%	D03
V03	DTP-containing vaccine	89%	D01

AdvisoryRule: (RuleId, RiskScore, EffectiveDate, RecommendationType, Notes, *CountryId, DiseaseId*) **Foreign Keys: CountryId, DiseaseId**

RuleId	RiskScore	EffectiveDate	Recommendation Type	Notes	Country Id	Disease Id

R01	5	01/01/2026	Required	Proof required for school/work entry in some regions	C01	D02
R02	4	01/01/2026	Recommended	Booster advised if last dose > 10 years	C01	D01
R03	3	01/01/2026	Recommended	Review immunity before traveling to crowded areas	C03	D03
R04	3	01/01/2026	Recommended	Increased risk in outbreaks; confirm vaccination	C02	D04

AdvisoryReport: (ReportId, SummaryNotes, GeneratedDate, ConsultationFee, *TripId*) **Foreign Key: TripId**

ReportId	SummaryNotes	GeneratedDate	ConsultationFee	TripId
AR01	Reviewed trip itinerary and vaccine history; generated recommendations	06/01/2024	75	TP001
AR02	Discussed routine immunizations and travel risk precautions	05/01/2024	60	TP002
AR03	Reviewed outbreak-related concerns and vaccination gaps	08/25/2024	80	TP003

ReportItem: (ReportItemId, RecommendationType, Reason, *ReportId, VaccineId*) **Foreign Keys: ReportId, VaccineId**

ReportItemId	RecommendationType	Reason	ReportId	VaccineId
RI01	Required	Measles risk in crowded travel settings; ensure immunity	AR01	V01

RI02	Recommended	Consider rubella protection based on history	AR01	V02
RI03	Recommended	Booster advised for diphtheria protection	AR02	V03
RI04	Recommended	Measles immunity recommended due to outbreak risk	AR03	V01